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HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING



CAPSTONE PROJECT 1 REPORT

TOPIC:

**DESIGN AND LAYOUT OF A PLC BOARD INTEGRATING AN STM32
MICROCONTROLLER USING ALTIUM SOFTWARE.**

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FACULTY OF ELECTRICAL AND ELECTRONICS ENGINEERING		CAPSTONE PROJECT N°1 ASSIGNMENT	
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INTRODUCTION

In modern industrial automation, Programmable Logic Controllers (PLCs) serve as the core processing units for production lines and machinery. While commercial PLCs are highly durable, they often suffer from high costs, proprietary ecosystems, and rigid peripheral configurations. This project addresses these limitations by designing a high-performance, custom PLC board integrating the STM32 microcontroller. Leveraging the functional pins of the STM32, including its advanced timing, multi-channel analog-to-digital converter (ADC), and hardware communication controller, this custom platform provides a highly adaptable hardware solution for diverse industrial control applications.

Implementing a standard microcontroller in an industrial environment requires addressing significant electrical challenges, such as electromagnetic interference, voltage surges, high-frequency noise, and ground loops. Without comprehensive hardware protection, peripheral signals can degrade, potentially causing permanent damage to the controller. Standard development setups do not meet these stringent requirements. Therefore, designing a reliable PLC board requires systematic hardware engineering at the schematic and PCB layout level. This includes implementing optical isolation for digital inputs, using power MOSFETs with short-circuit protection for digital outputs, integrating precise current-to-voltage conversion circuits with Zener cutoff circuits for similar safety, and implementing dedicated transceivers for robust CAN-bus communication.

To solve these challenges, this project focuses on the Design and Layout of a PLC Board Integrating an STM32 Microcontroller using Altium Software. The primary objective is to develop a comprehensive schematic and an optimized, production-ready PCB layout compliant with industrial electrical standards. The scope encompasses the engineering of robust power distribution, multi-channel opt isolated digital I/O, signal-conditioned analog stages (including 4-20mA monitoring and DAC output drive configurations), high-speed input capture/encoder interfaces, and a protected CAN-bus interface. Altium Designer is utilized to optimize signal integrity, implement localized

ground planes, and enforce proper physical clearance, ensuring stable and reliable hardware operation.

I would like to express my sincerest gratitude to my supervisor, MSc Bui Quoc Bao, for dedicating his time to guiding and providing weekly feedback throughout the project, which enabled me to achieve the best possible results.

CHAPTER 1: TOPIC OVERVIEW

1.1. Reasons for Choosing the Topic

The decision to design a custom STM32 integrated PLC board stemmed from the need to overcome the rigid configuration and high cost of commercial PLCs. While standard industrial controllers are reliable, they lack the flexibility necessary for specialized applications requiring specific peripheral combinations within a compact footprint. By directly embedding the high-performance STM32 microcontroller into the hardware, this project created a highly adaptable and cost-effective platform capable of advanced signal processing. This approach allows for the integration of specialized industrial interfaces—such as optically isolated inputs, high-current MOSFET outputs, precise 4-20mA analog signal regulation, and CAN-bus communication—perfectly customized for specific operational needs without redundant hardware.

Furthermore, deploying a sensitive microcontroller in a harsh industrial environment presents significant engineering challenges, particularly concerning electromagnetic interference (EMI) and high voltage surges. This project provides essential research on hardware resilience and robust board layout design using Altium Designer software. The main focus is on implementing robust protection circuit strategies, including optocoupler isolation, Zener diode signal clipping, and self-resetting fuses to ensure signal integrity and protect the control core from electrical damage. The ultimate goal is to select this topic to master industrial-grade Electronic Design Automation (EDA), bridging the gap between theoretical schematics and production-ready PCB layouts, and providing interference resistance.

1.2. Project objectives

Learn about the basic components in a PLC circuit and further improve other components.

Study and become familiar with Altium Designer for schematic capture, PCB layout design, and its various other features.

1.3. Scope of project

This project focuses on three parts:

- Design the schematic and analyze the functional blocks.
- Simulate the operation of the functional blocks.
- Design the PCB layout and generate the manufacturing output files.

CHAPTER 2: SCHEMATIC DESIGN AND OPERATION

2.1. Project Structure

The figure below illustrates the Project Structure of the STM32-PLC system inside Altium Designer. This tree view manages all the essential source documents required for the hardware design, which includes the schematic sheets from the cover page, power blocks, and core MCU to the isolated I/O interfaces, the PCB layout document (PCB1.PcbDoc), and the Bill of Materials (DA1_prj.BomDoc).

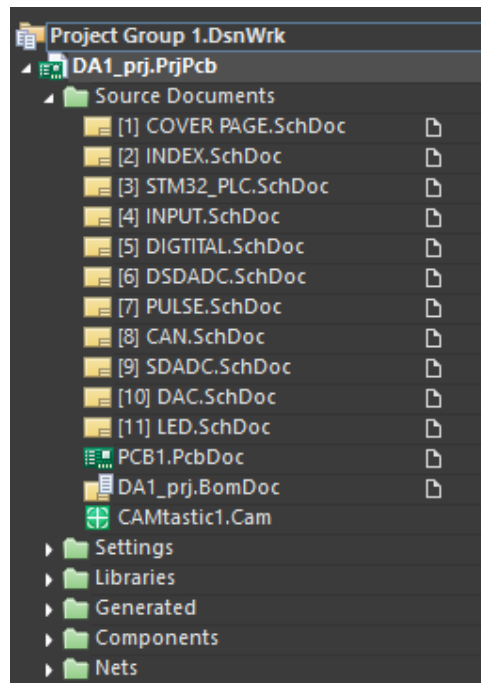


Image 2.1. Project panel in Altium software

There are 9 most important files that need to be focused on to understand and evaluate the layout and operation of the board, the detail schematic of each block will be illustrated in next part.

STM32-PLC (Central Unit), It includes the STM32F373VBTx central microcontroller (QFP100 package), MIC5504-3.3YM5 voltage regulator IC, HSE and LSE external crystals, 3002TR current pins and pin socket, and connectors supporting ST-Link V2 programmer interface.

INPUT block, includes PC817 optocoupler ICs, a 10-position DIP switch, and a system of strip resistors and surface-mount resistors (SMDs) for current limiting.

DIGITAL block, comprising N-channel MOSFET power transistors (BUK9M15-60EX, surface-mount package LFPAK33), combined with Polyfused self-resetting fuses at each output.

DSDAC block, includes precision resistors with a value of 160 Ohms, Polyfused self-resetting fuses, and a system of BZX84C3V6 Zener diodes for protection.

PULSE block, includes SN74HC14N integrated Schmitt-Trigger logic ICs, combined with PC817 opto-isolator ICs and passive components (resistors, capacitors) to form a filter circuit.

CAN block, includes the TCAN332D dedicated CAN network transceiver IC, transmission noise filtering capacitors, and a diode protection system against high voltage surges.

SDADC block, includes 160 Ohm precision resistors, Polyfused self-resetting fuses, and Zener diodes for analog input protection.

DAC block, includes operational amplifier (Op-amp) ICs LM2904DR and LM2902, combined with BC847C inverse transistors and surface-mount resistors.

LED block, includes 220 Ohm current-limiting resistors combined with surface-mount SMD (SMD) LEDs: LED_USB_RX_TX, LED_CAN_RX_TX, and LED_ERROR.

Beside that, there are lots of “Footprint Library” folder that are used in this project, since there are too many of that folder, the Footprint of each component will not be listed in this project.

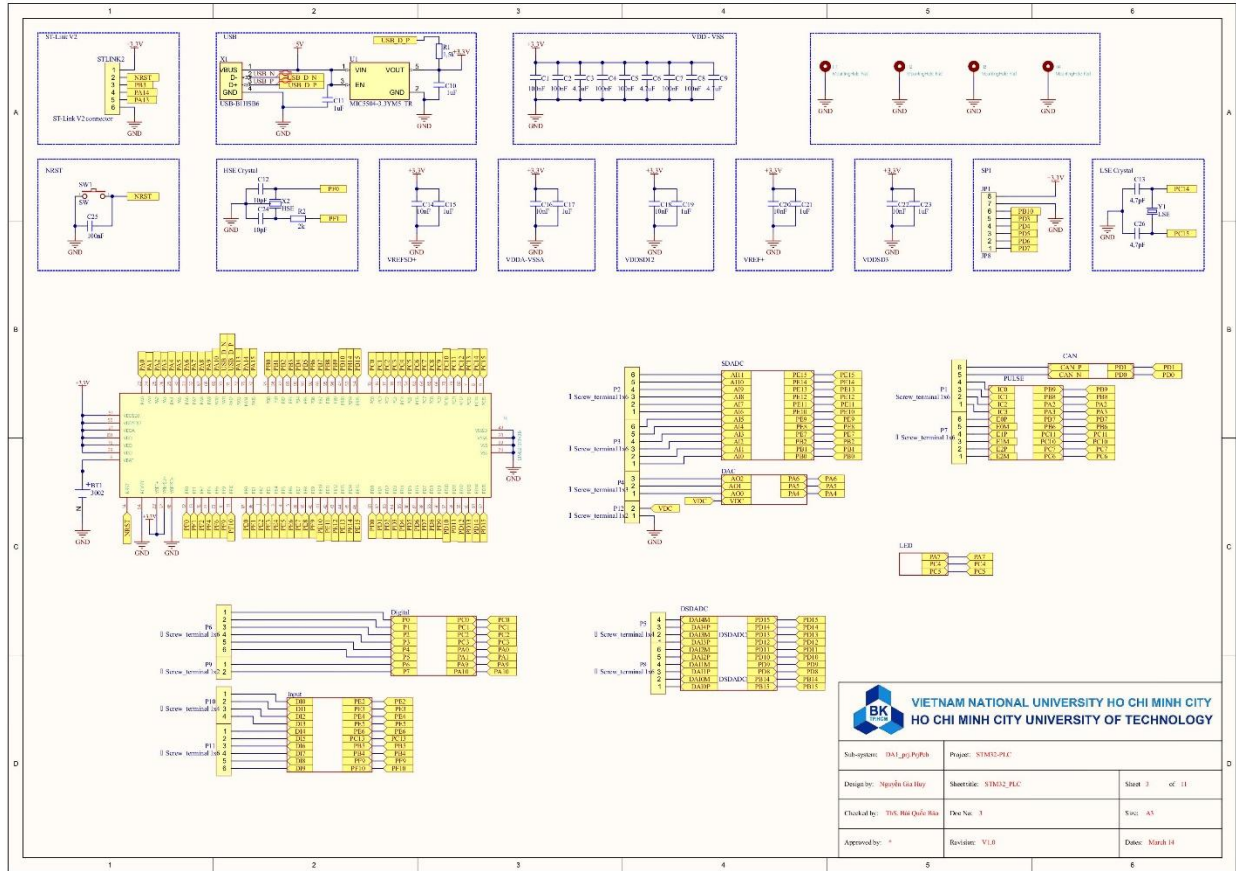


Image 2.2. First-look of the system

2.2. Design and Simulation result of each part

2.2.1. INPUT (DIGITAL INPUT)

Name	I/O	Max V	Min V	Operation
DI 9	INPUT	30V	-1V	Measuring 0 to 1
DI 8	INPUT	30V	-1V	Measuring 0 to 1
DI 7	INPUT	30V	-1V	Measuring 0 to 1
DI 6	INPUT	30V	-1V	Measuring 0 to 1
DI 5	INPUT	30V	-1V	Measuring 0 to 1
DI 4	INPUT	30V	-1V	Measuring 0 to 1
DI 3	INPUT	30V	-1V	Measuring 0 to 1
DI 2	INPUT	30V	-1V	Measuring 0 to 1
DI 1	INPUT	30V	-1V	Measuring 0 to 1
DI 0	INPUT	30V	-1V	Measuring 0 to 1

Table 2.1. INPUT pinouts and properties

Generally, the microcontroller safely receives On/Off signals (logic level 0 or 1, or industrial voltage such as 24V) from the external digital device thanks to its electrical isolation feature using an optocoupler. The PC817 optocoupler is an optocoupler IC that completely isolates the 24V industrial power supply voltage at external devices from the low 3.3V voltage of the microcontroller, protecting the MCU from burnout in case of overvoltage.

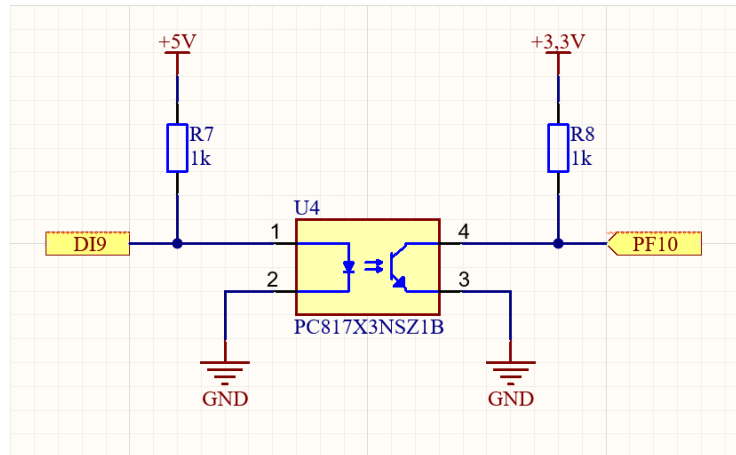


Image 2.3. Input circuit using optocoupler

The DIP switch in the INPUT block is used to determine whether that Screw Terminal pin is active or whether the STM32 microcontroller can read a digital value at that pin.

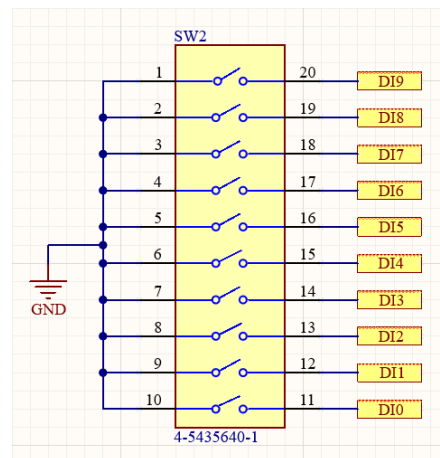


Image 2.4. 10-position DIP Switch

For simulation, because I need to perform a simulation with digital input, so I can use Proteus to run the simulation results.

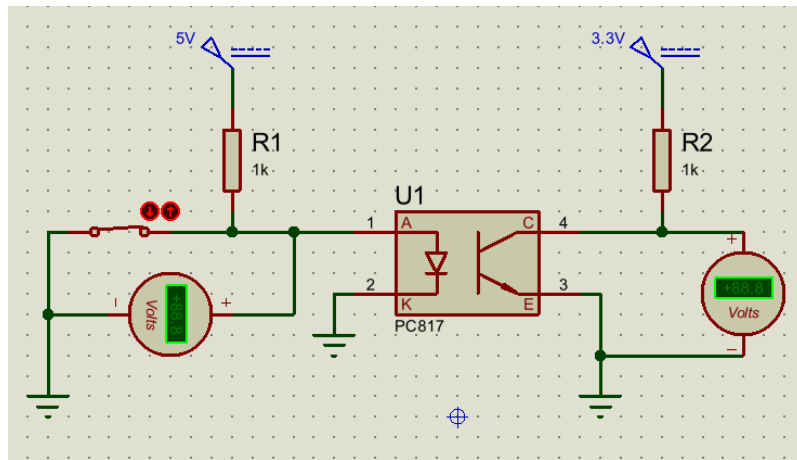


Image 2.5. The circuit diagram simulates an input circuit using an input signal isolation power supply.

Firstly, when the switch is open (low level), the output voltage of the optocoupler supplied to the internal power supply of the MCU will be 0V.

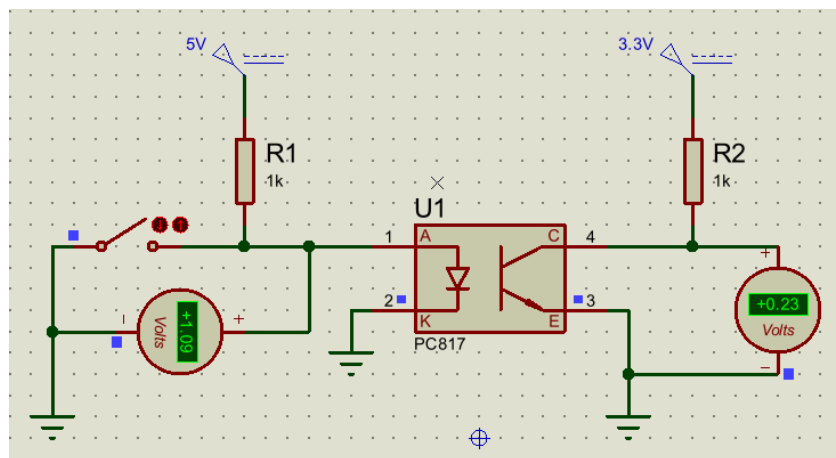


Image 2.6. In the case where the switch is opened

Next, when the switch is closed (high), the output voltage of the optocoupler supplied to the internal power supply of the MCU will be 3.3V.

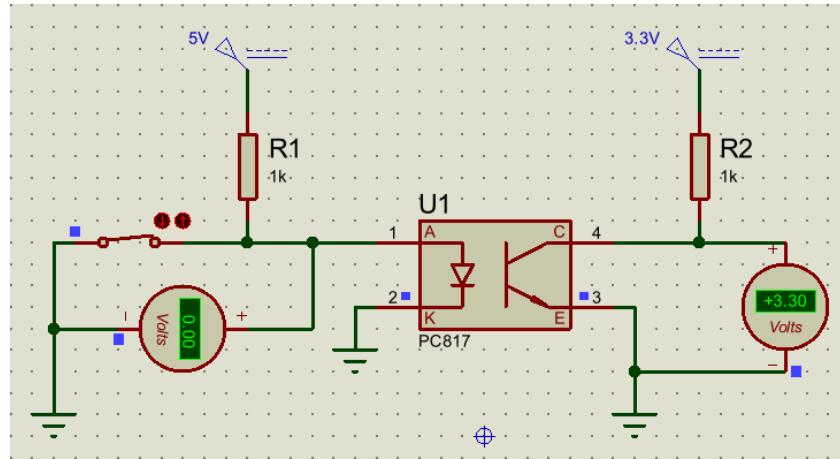


Image 2.7. In the case where the switch is closed

2.2.2. DIGITAL OUTPUT (PWM)

Name	I/O	Max V	Min V	Operation
PWM 7	INPUT (open drain)	30V	-1V	N-ch MOSFET PWM open drain
PWM 6	INPUT (open drain)	30V	-1V	N-ch MOSFET PWM open drain
PWM 5	INPUT (open drain)	30V	-1V	N-ch MOSFET PWM open drain
PWM 4	INPUT (open drain)	30V	-1V	N-ch MOSFET PWM open drain
PWM 3	INPUT (open drain)	30V	-1V	N-ch MOSFET PWM open drain
PWM 2	INPUT (open drain)	30V	-1V	N-ch MOSFET PWM open drain
PWM 1	INPUT (open drain)	30V	-1V	N-ch MOSFET PWM open drain
PWM 0	INPUT (open drain)	30V	-1V	N-ch MOSFET PWM open drain

Table 2.2. DIGITAL OUTPUT pinouts and properties

This block functions to output control signals from the microcontroller to operate external actuators.

The BUK9M15-60EX MOSFET acts as a high-speed electronic relay, receiving trigger signals from the IO pin of the STM32 to switch the power supply to the load on/off.

The Pollyfuse provides overcurrent and short-circuit protection. When an external load (such as a solenoid valve, motor, lamp, etc.) fails or the wires short-circuit, causing a surge in current, this fuse will automatically increase its internal resistance to an extremely high level to interrupt the current, protecting the MOSFET from burning out. Unlike traditional lead-wire fuses, the Polyfuse, once cooled and after the external short circuit

has been resolved, will automatically return to its normal conductive state, allowing the machine to continue running.

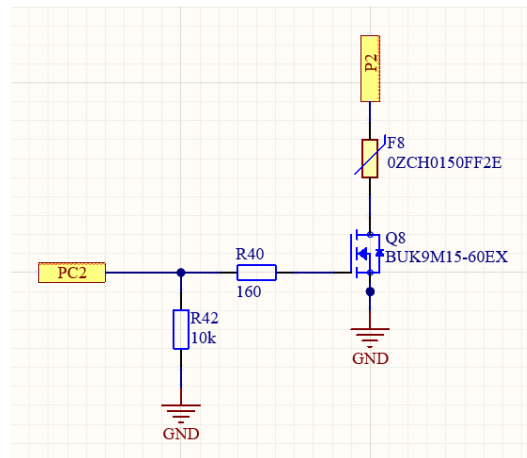


Image 2.8. Digital output circuit with MOSFET and Polyfuse

2.2.3. DSDADC - Differential Sigma-Delta Analog to Digital Converter

Name	I/O	Max V	Min V	Operation
DADC 4 -	Input	30V	-1V	Measuring -32768 to 32767
DADC 4 +	Input	30V	-1V	Measuring -32768 to 32767
DADC 3 -	Input	30V	-1V	Measuring -32768 to 32767
DADC 3 +	Input	30V	-1V	Measuring -32768 to 32767
DADC 2 -	Input	30V	-1V	Measuring -32768 to 32767
DADC 2 +	Input	30V	-1V	Measuring -32768 to 32767
DADC 1 -	Input	30V	-1V	Measuring -32768 to 32767
DADC 1 +	Input	30V	-1V	Measuring -32768 to 32767
DADC 0 -	Input	30V	-1V	Measuring -32768 to 32767
DADC 0 +	Input	30V	-1V	Measuring -32768 to 32767

Table 2.3. DSDADC pinouts and properties

The circuit receives analog signals (4-20mA current) from sensors, then passes through a 160 Ohm resistor to convert it into an oscillating voltage in the range of 0.64 to 3.2V, a voltage level that the STM32 microcontroller can recognize. A diode functions to ground the voltage if a sudden voltage surge is detected and reverse current or open circuit occurs.

The DSDADC block receives input signals from the two DADC+ and DADC- terminals, typically used for high-precision sensors in environments with significant noise.

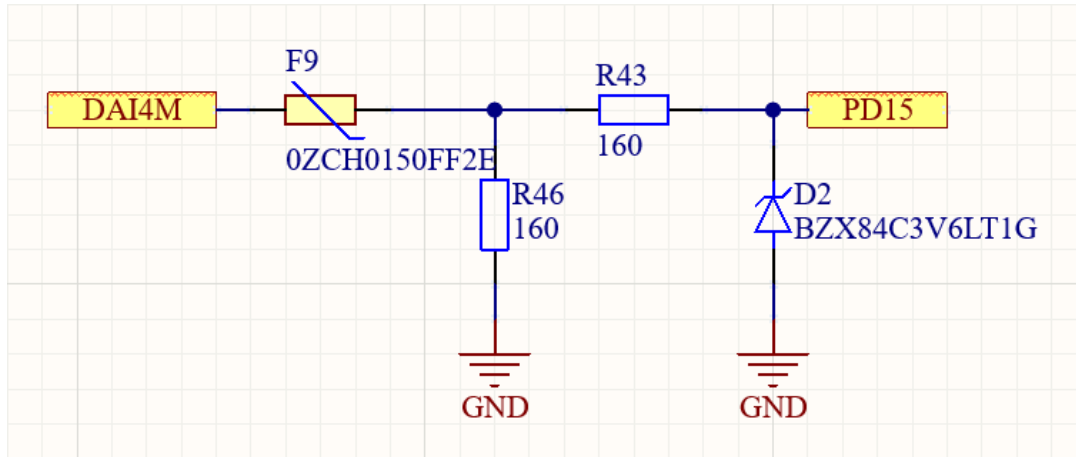


Image 2.9. DSDADC circuit

For the simulation, I chose to perform it on LTspice since it required an analog input.

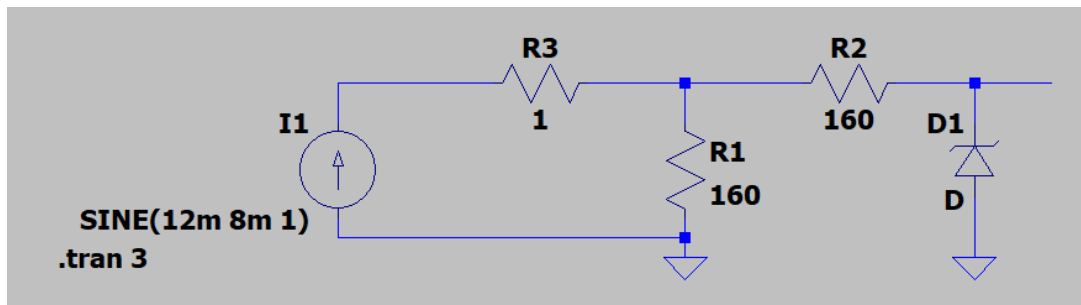


Image 2.10. DSDADC circuit diagram

A 4-20mA sine wave signal is passed through a fuse (since the LTspice component library doesn't include a fuse, a small resistor is used instead to measure the current flowing through the load).

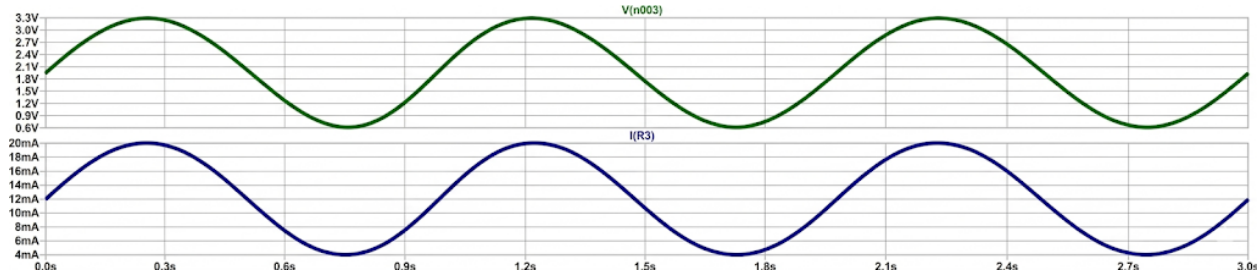


Image 2.11. DSDADC simulation result

When a 4-20mA current is passed through R3 (Fuse), the output voltage is observed to be in the range of 0.64-3.2V.

Verification using Ohm's Law:

+ Ohm's Law: $U = I \times R$

+ Assuming $I = 4\text{mA}$: $U = 4 \times 160 = 640\text{mV} = 0.64\text{V}$

+ Assuming $I = 20\text{mA}$: $U = 20 \times 160 = 3200\text{mV} = 3.2\text{V}$

2.2.4. SDADC - Single-Ended Sigma-Delta Analog to Digital Converter

Name	I/O	Max V	Min V	Operation
ADC 11	Input	30V	-1V	Measuring from 0 to 65535
ADC 10	Input	30V	-1V	Measuring from 0 to 65535
ADC 9	Input	30V	-1V	Measuring from 0 to 65535
ADC 8	Input	30V	-1V	Measuring from 0 to 65535
ADC 7	Input	30V	-1V	Measuring from 0 to 65535
ADC 6	Input	30V	-1V	Measuring from 0 to 65535
ADC 5	Input	30V	-1V	Measuring from 0 to 65535
ADC 4	Input	30V	-1V	Measuring from 0 to 65535
ADC 3	Input	30V	-1V	Measuring from 0 to 65535
ADC 2	Input	30V	-1V	Measuring from 0 to 65535
ADC 1	Input	30V	-1V	Measuring from 0 to 65535
ADC 0	Input	30V	-1V	Measuring from 0 to 65535

Table 2.4. SDADC pinouts and properties

The SDADC receives an analog signal in the form of a 4-20mA current from sensors, then passes through a 160 Ohm resistor to convert it into an oscillating voltage in the range of 0.64 to 3.2V, a voltage level that the STM32 microcontroller can recognize. A

diode functions to ground the voltage if a sudden voltage surge or reverse current/open circuit is detected.

Unlike DSDADCs, SDADCs typically only receive input signals directly from peripheral devices and common sensors at close range.

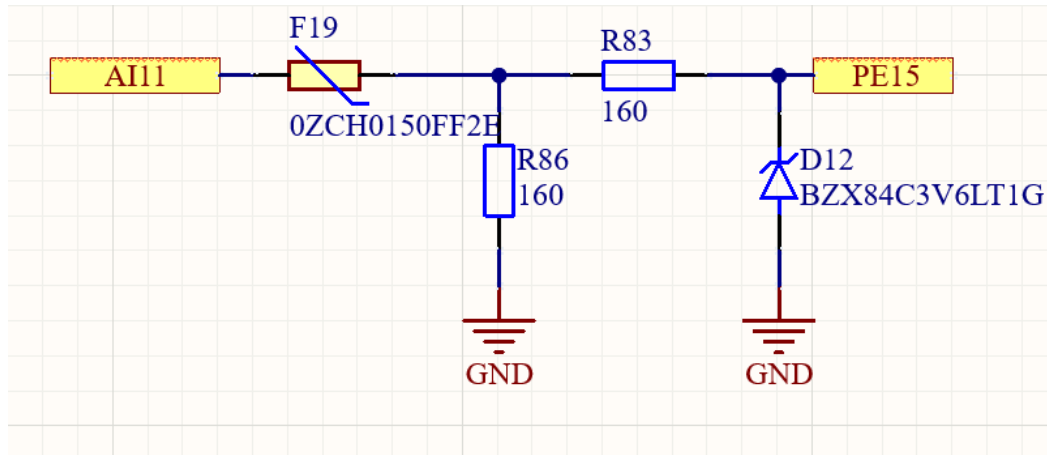


Image 2.12. SDADC circuit

For simulation, here I also use LTspice to perform simulation since it require analog inputs.

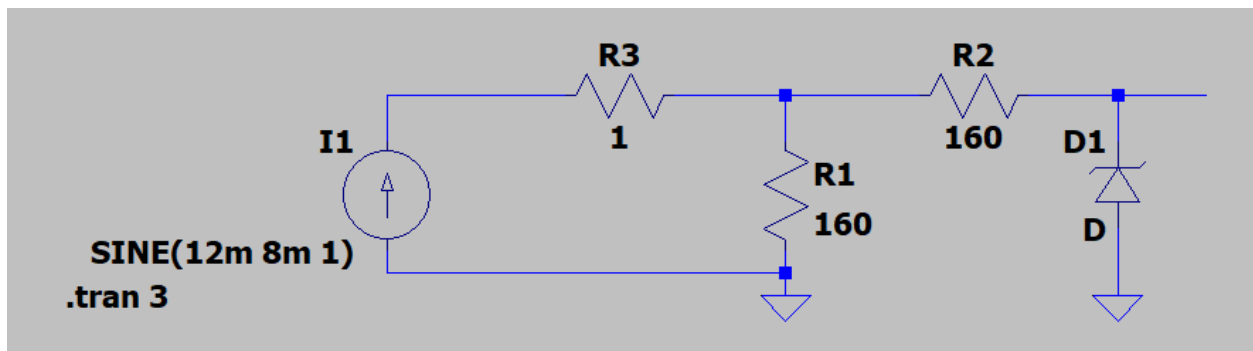


Image 2.13. SDADC circuit diagram

A 4-20mA sine wave signal is passed through a fuse (since the LTspice component library doesn't include a fuse, I replaced it with a small resistor to measure the current flowing through the load).

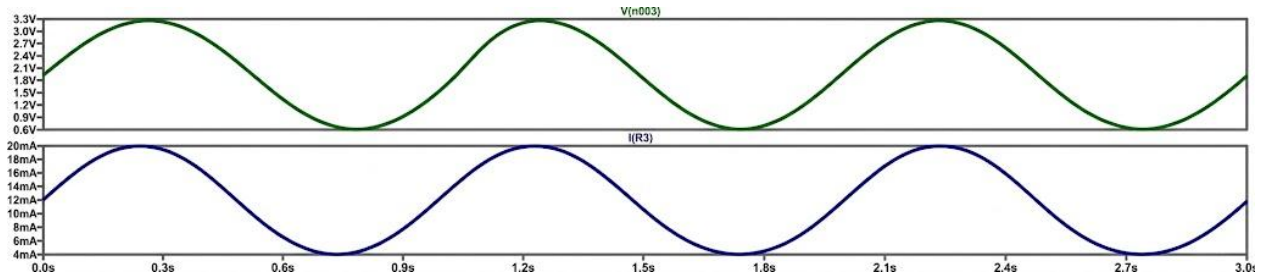


Image 2.14. SDADC simulation result

2.2.5. DAC – Digital to Analog Converter

Name	I/O	Max V	Min V	Operation
AO2	Output	30V	-1V	0–20 mA output
AO1	Output	30V	-1V	0–20 mA output
AO0	Output	30V	-1V	0–20 mA output
VDC	Input	32V	-1V	Power supply for 0–20 mA output
GND	Output	0V	0V	Ground

Table 2.5. DAC pinouts and properties

The three DAC outputs provide a 4-20 mA current loop output with 12-bit resolution.

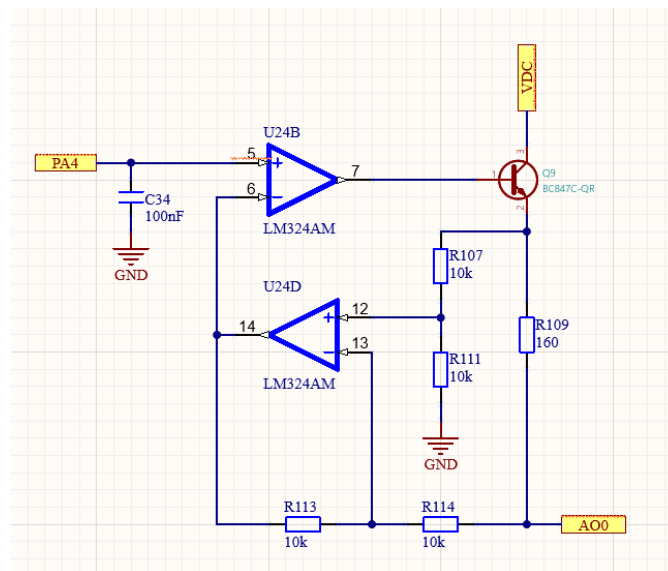


Image 2.15. DAC circuit

The DAC block uses an internal STM32 converter combined with an op-amp and transistor amplifier circuit to convert voltage into a 4-20 mA current signal. This design

allows the PLC to precisely control devices remotely with 12-bit resolution, while also ensuring noise immunity and detection of broken wires in the system.

The simulation is performed using LTspice.

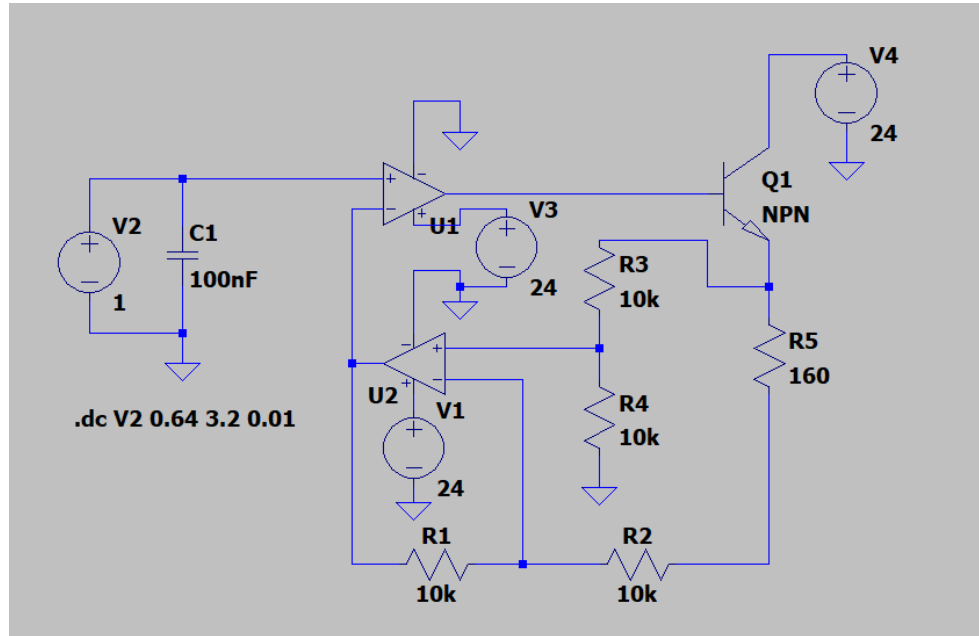


Image 2.16. DAC circuit diagram

Put a voltage signal that increases continuously from 0.64 - 3.2V (Similar to the signal that STM32 can output) into a 100nF signal filter capacitor and then enter the first Op-amp amplifier circuit.

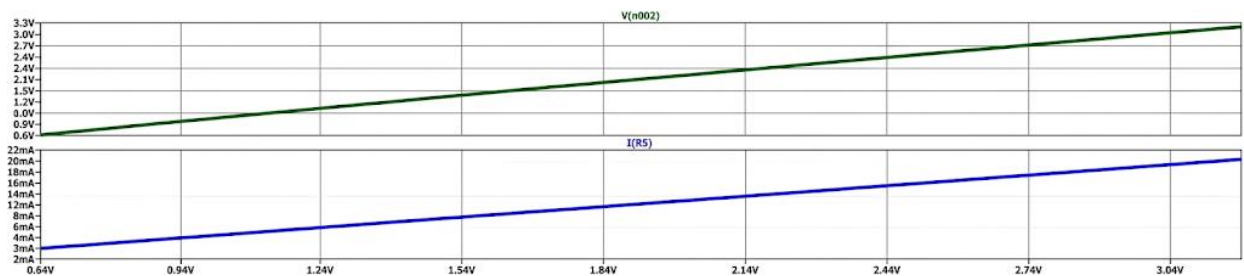


Image 2.17. DAC simulation result

Observe the input signal (above), which is a voltage of 0.6 – 3.2V, and it will be converted into a current signal (below) of 4 – 20mA.

2.2.6. PULSE (INPUT CAPTURE / ENCODER)

2.2.6.a. Input capture

Name	I/O	Max V	Min V	Operation
IC 0	Input	30V	-1V	Frequency 0–10 kHz
IC 1	Input	30V	-1V	Frequency 0–10 kHz
IC 2	Input	30V	-1V	Frequency 0–10 kHz
IC 3	Input	30V	-1V	Frequency 0–10 kHz

Table 2.6. Input Capture pinouts and properties

Input capture is used to measure the frequency of a signal from an external sensor. It also uses optocouplers to isolate and protect the microcontroller from overvoltage issues caused by the STM32. The SN74HC14N inverter at the end is used as a Schmitt trigger to make the output signals become "sharpen".

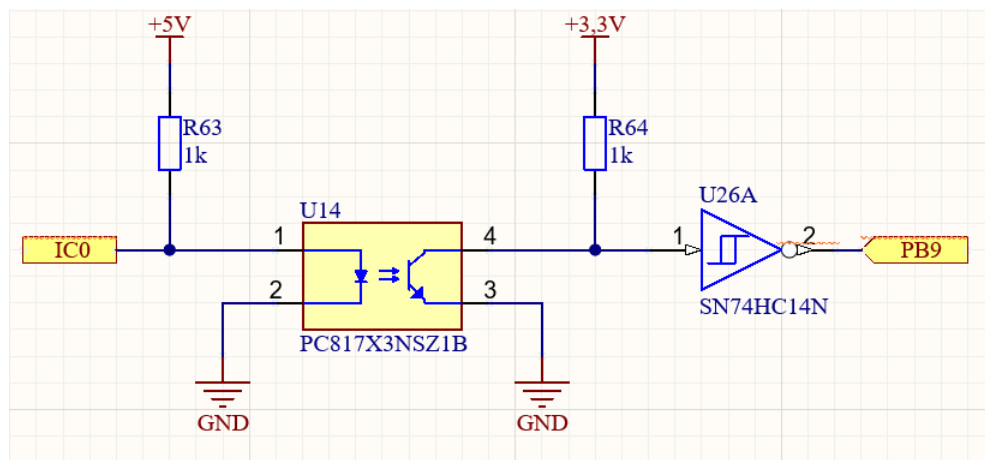


Image 2.18. Input Capture circuit

2.2.6.b. ENCODER

Name	I/O	Max V	Min V	Operation
Encoder 0 +	Input	30V	-1V	Counting -32768 to 32767
Encoder 0 -	Input	30V	-1V	Counting -32768 to 32767
Encoder 1 +	Input	30V	-1V	Counting -32768 to 32767
Encoder 1 -	Input	30V	-1V	Counting -32768 to 32767
Encoder 2 +	Input	30V	-1V	Counting -32768 to 32767
Encoder 2 -	Input	30V	-1V	Counting -32768 to 32767

Table 2.7. ENCODER pinouts and properties

ENCODERS are designed to count rotations and determine the position (forward or backward) of machinery and motors using two pulse channels.

Same as INPUT CAPTURE, the ENCODER also use a PC817X optocoupler isolated power supply for microcontroller safety protection and an inverter at the end to regulate the signal. However, unlike the INPUT CAPTURE block, the ENCODER block receives signals from both Encoder+ and Encoder- terminals 24V and converts them completely to a 3.3V logic level via an optocoupler.

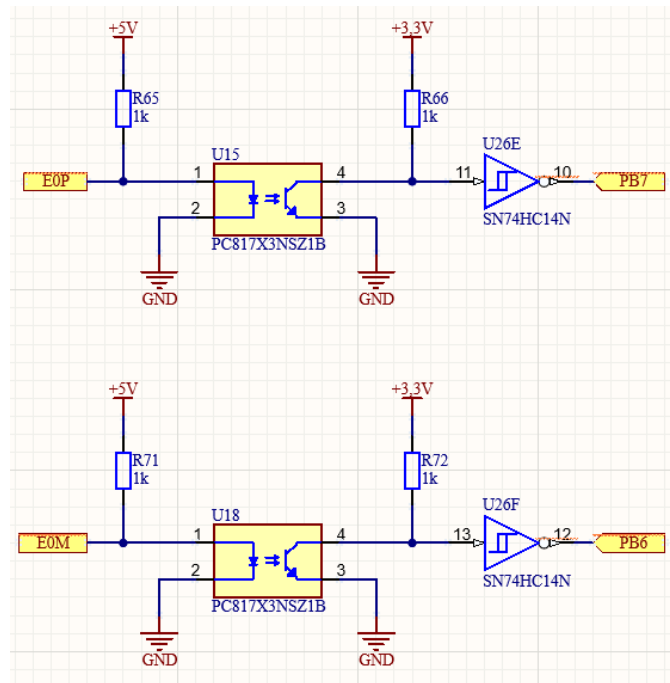


Image 2.19. ENCODER circuit

2.2.6.c. Simulation using LTspice

Since the INPUT CAPTURE and the ENCODER have the same circuit diagram, we will use the same simulation result for both of them.

First, we will take a look at the case where the circuit is not using the Inverter SN74HC14N at the output.

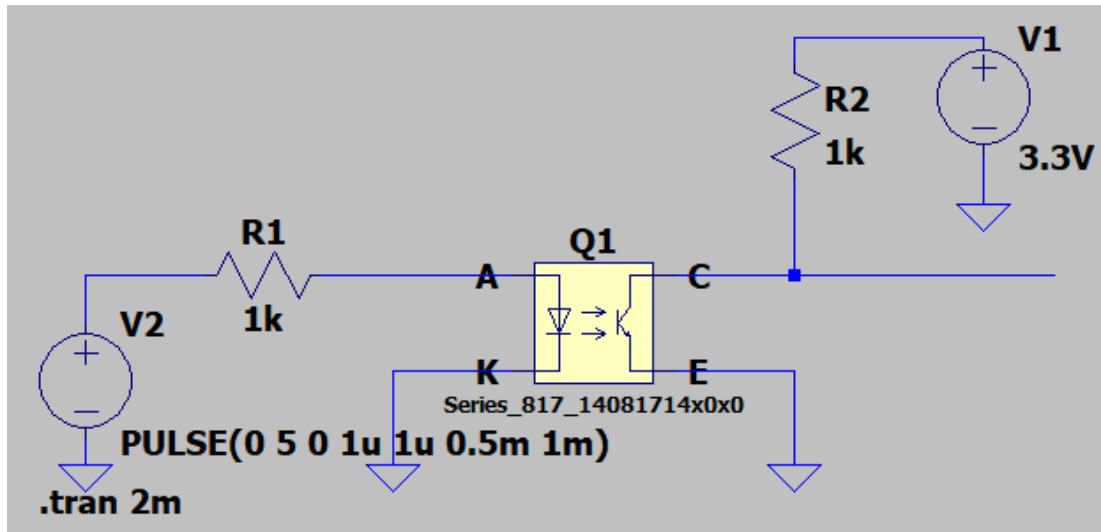


Image 2.20. PULSE circuit diagram without Inverter

Given a "Pulse" input signal with the adjusted parameters as shown in the image, the input pulse will range from approximately 0-5V.

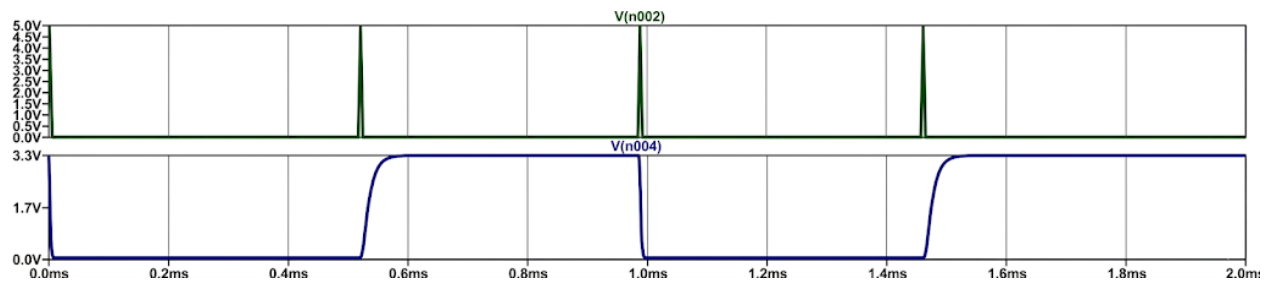


Image 2.21. PULSE simulation results (without Inverter)

Because the input signal is too large compared to the MCU's voltage, it will be reduced to a voltage level that the MCU can receive (0-3.3V) through the Input Capture block.

Observing the simulation results of the output pulse, it was found that the output signal was slightly "curved" compared to the input signal, so I passed the output through a logic gate with a Schmitt Trigger function to make the pulse square again.

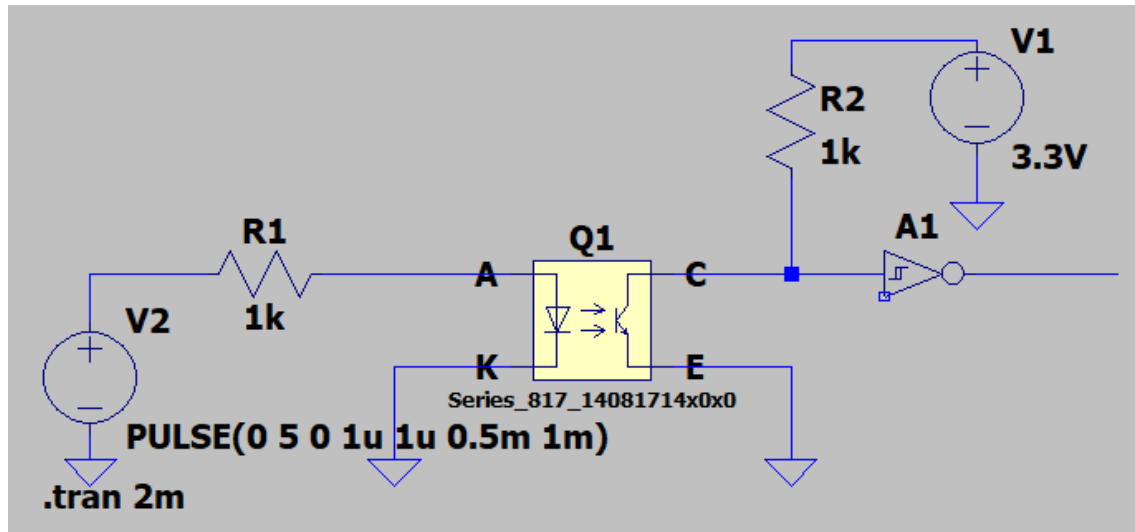


Image 2.22. PULSE circuit diagram with an Inverter

Keep the same simulation set up, when we run the simulation again, we can observe that the after result is different than the previous one.

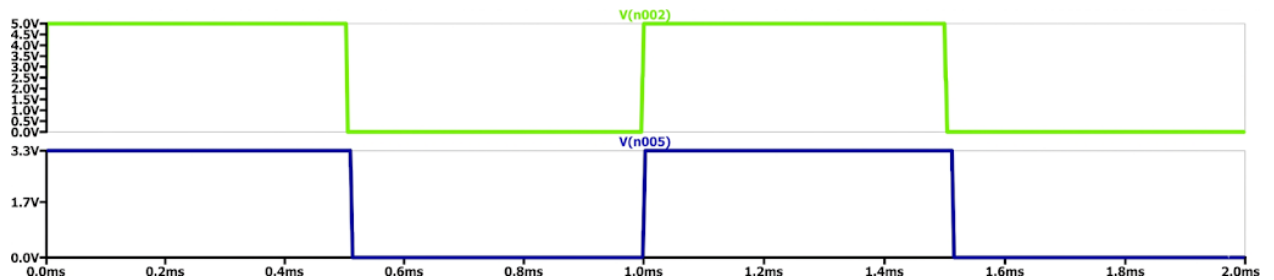


Image 2.23. PULSE simulation results (with an Inverter)

2.2.7. CAN

Name	I/O	Max V	Min V	Operation
CAN High	Both	14V	-15V	CAN communication
CAN Low	Both	14V	-15V	CAN communication

Table 2.8. CAN pinouts and properties

The CAN bus block uses the TCAN332 transceiver IC to enable PLC communication via high-speed differential standard. The integrated circuit includes a Split Termination filter (120Ω) to prevent signal reflection and a PESD5V0 diode for electrostatic protection, ensuring data integrity in environments with strong electromagnetic interference.

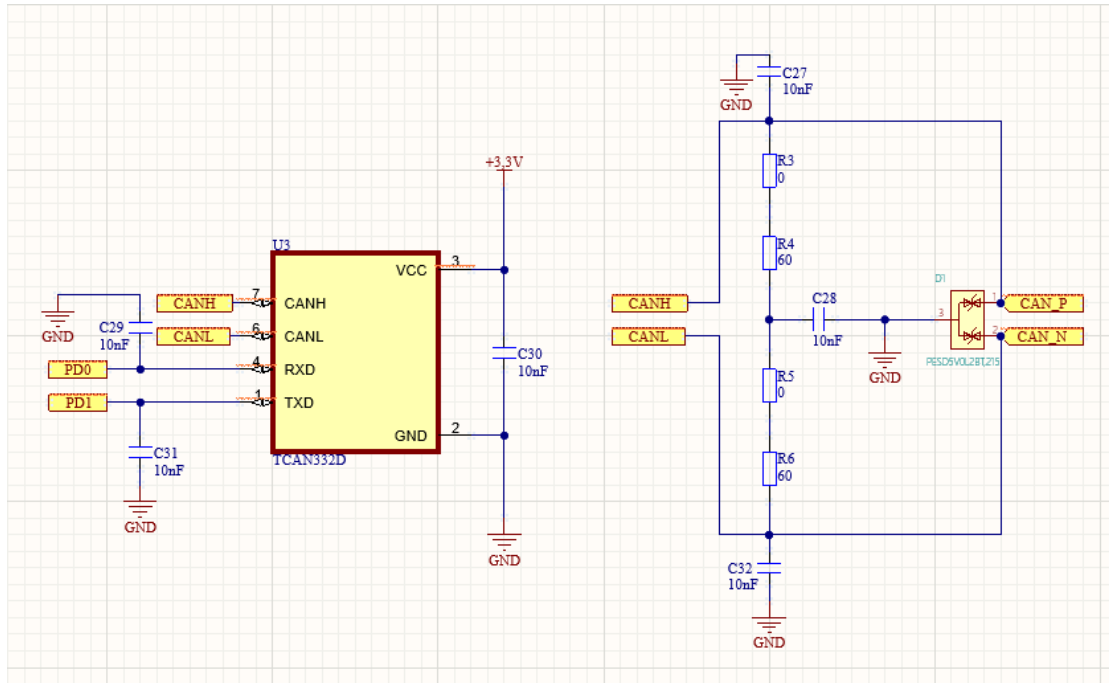


Image 2.24. CAN circuit

CHAPTER 3: PCB LAYOUT AND HEALTH CHECK

3.1. PCB layout using Altium designer

This project used a total of 4 layer that are named F.Cu, In1.Cu, In2.Cu, B.Cu respectively.

First layer – F.Cu (Top layer), this is where most SMD components (such as STM32 chips, 74HC14N ICs, optocouplers, MOSFETs, etc.) are located. This layer is used for high-speed logic signal paths and short circuits connecting components. The remaining empty space is usually filled with copper and connected to the GND network to cool components (such as MOSFETs) and create a loop noise shield.

Second layer - In1.Cu (Inner Layer 1), this layer is configured as a flat, continuous plane containing only copper connections to the GROUND (GND) pin, with no signal wires crossing it (or very limited signal wires).

Third layer - In2.Cu (Inner Layer 2), this layer is dedicated to distributing the power supply to the microcontroller (+3.3V). Instead of having to run tangled, complicated power wires at the top layer that clutter the circuit, the 3.3V power supply is now a network covering the entire layer 3.

Fourth layer - B.Cu (Bottom Layer), used to route remaining signal lines that could not be fully routed in the top layer. The remaining area was also filled with GND copper.

Name	Area	Net
[L1] Top		
F.Cu-GND	11067.829 sq.mm	GND
[L2] GND		
In1.Cu-GND	17303.824 sq.mm	GND
[L3] POWER		
In2.Cu-+3V3	15631.617 sq.mm	+3,3V
[L4] BOTTOM		
B.Cu-GND	16167.916 sq.mm	GND

Image 3.1. Polygon manager

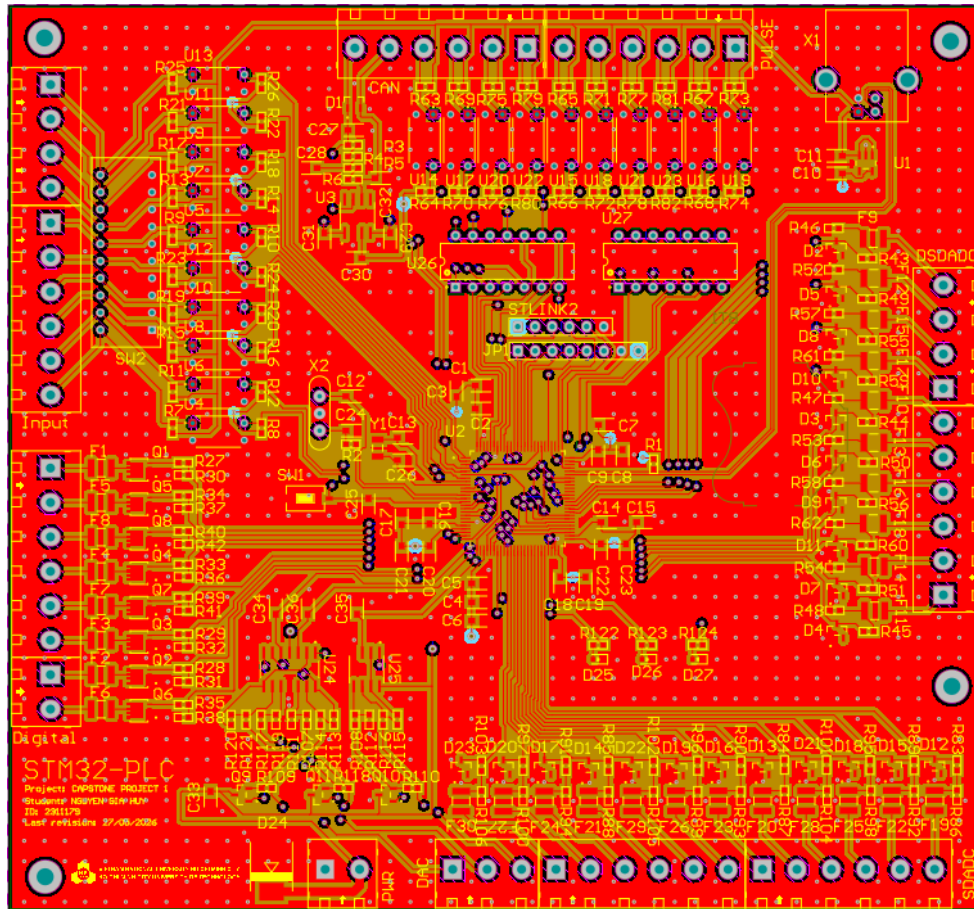


Image 3.2. PCB layout in 2D mode

First, regarding component placement, the screw terminals have been positioned close to the edges and near the keep-out layer of the board for easy connection to other peripherals without obstruction. The decoupling capacitors have been placed as close as possible to the STM32 microcontroller, and the resistors and capacitors from other components are also placed in the correct order according to the signal path as shown in the schematic.

We can observe that the INPUT and DIGITAL blocks are located on the left side of the PCB, CAN and PULSE blocks share a same Screw Terminal on the top of the PCB, also, DAC and SDADC blocks also use one Screw Terminal and is located at the bottom of the PCB, lastly, the DSDADC blocks is standing alone on the right side of the PCB. All

components within the blocks have been arranged in the correct order and linked as shown in the schematic.

For the routing work, most wires are routed in the first layer. In cases of insufficient space, cross-connection, etc., they may be connected via a via to the fourth layer and then back to the first layer via a via with the same network name. The wire width is specified from 0.2mm to 10mm, with 0.254mm being commonly used. In addition to standard signal wires, this circuit also uses high-speed, differential-pair cables for USB and CAN, ensuring equal cable length and signal transmission speed for both channels, thus preventing errors.

In addition to vias used to connect wires between two layers, stitching vias have been added automatically to ensure a good return path and also act as fan outputs to cool the circuit, preventing overheating and potential fire hazards.

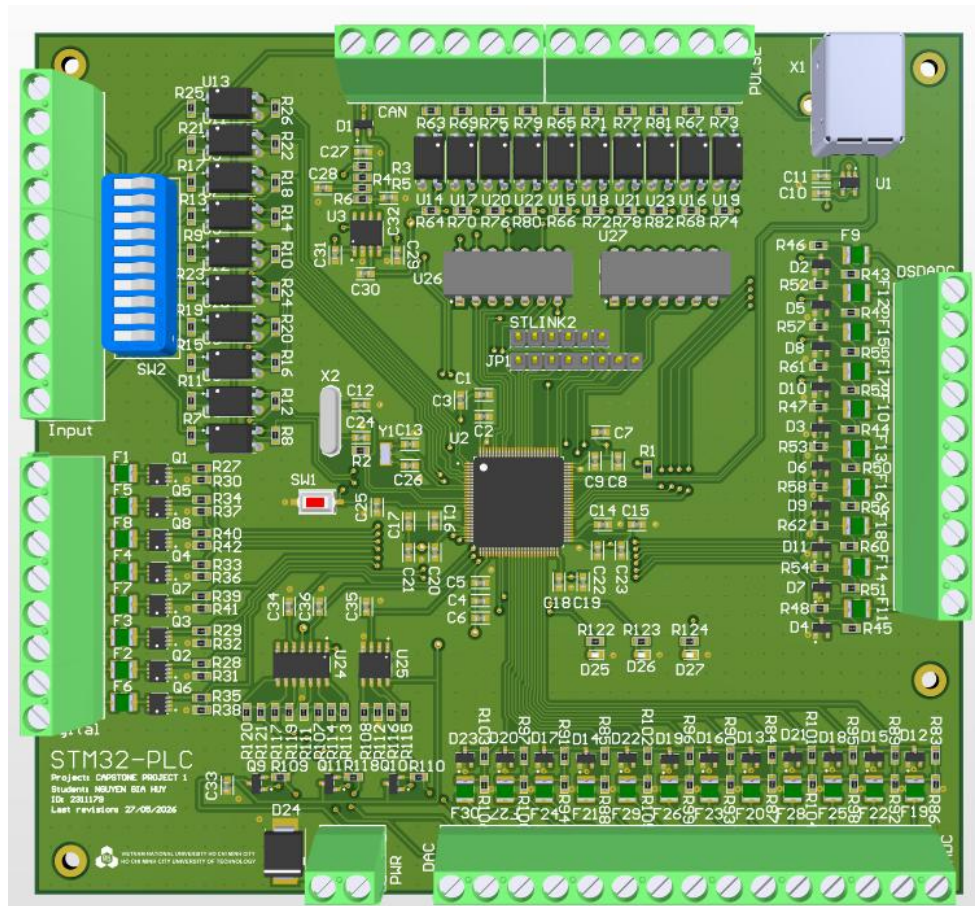


Image 3.3. PCB layout in 3D mode (front)

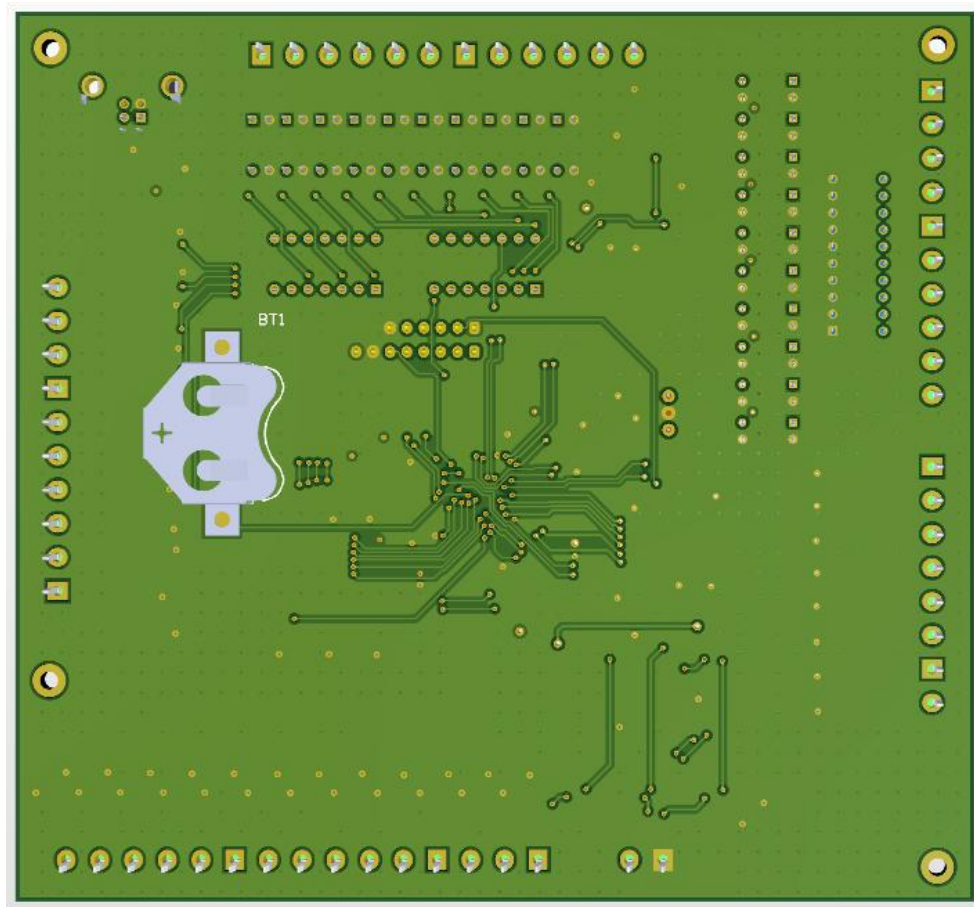


Image 3.4. PCB layout in 3D mode (back)

With the results from DRC, we can at least be sure that this design is not flawed in terms of the rules we defined earlier. It also helps to ensure that all wires and components are connected, with no short circuits or overlooked components.



Design Rule Verification Report

Date:	07/06/2026	Warnings:	0
Time:	05:57:54	Rule Violations:	0
Elapsed Time:	00:00:01		
Filename:	C:\Users\nguyen\OneDrive\Máy tính\ALTIUM-PROJECT\3_DA1_prj\PCB1.PcbDoc		

Image 3.5. DRC result

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3. M. Faheemuddin, “*Decoupling Capacitor Placement Guidelines Every PCB Designer Should Know*,” ProtoExpress (Sierra Circuits), Oct. 7, 2025. Available: <https://www.protoexpress.com/blog/decoupling-capacitor-placement-guidelines-pcb-design/>
4. *Hướng dẫn Altium – Bản cập nhật 24 Oct*, Tài liệu hướng dẫn thiết kế PCB bằng Altium Designer, PDF.